

Data Mining - Fall 2026 - Group Project

Milestone 0: Proposal Submission

Group Names: Zach Erdley, Dylan Hudson, Mar Lonsway

1. Problem and Project Goal

Problem statement: Drug shortages are a constant problem for hospitals, pharmacies, and patients. Shortages can affect anything from over-the-counter medications to critical preparations for surgeries and emergency care. Warning times for shortages can be highly variable and unpredictable, making stockpiling and preparation difficult and expensive.

Primary question: *Can drug shortages be predicted?* Specifically, can we use manufacturer/supplier data, shortage history, and national use data to predict whether or not a certain drug will go into shortage within a particular time window?

Supporting questions: (a) Do manufacturers quietly exit a product, measurable as package codes being delisted, before a shortage is declared, and how far ahead? (b) Which drug characteristics raise risk most?

Why it matters: Vizient¹ estimates shortages cost hospitals ~\$900 million per year in labor costs alone, not counting price increases to patients and the health effects of being unable to obtain medication. Patients, hospitals, and pharmacies benefit most directly.

Expected evidence: A continuous score reflecting the probability that a drug will go into shortage within a particular time frame, which can then be calibrated and thresholded to a binary outcome if necessary. The level we will work on is by each active ingredient in each route of administration (2,897 drugs) and validate by training on earlier years and testing on later ones.

Data Source(s) and Access

Dataset	Provider	URL	Size	Variables
FDA Drug Shortages	openFDA	download.open.fda.gov/drug/shortages/	1634 recs	Status, posting date, package NDC, dosage form, company, reason
Archived FDA shortage files	Internet archive	web.archive.org -> accessdata.fda.gov/scripts/drugshortages/	102 snapshots, 2019–26	Shortage start and end dates, status, reason
NSDE (product lifecycle)	openFDA	download.open.fda.gov/other/nsde/	667457 recs	Marketing start/end, inactivation and reactivation dates
FDA NDC Directory	openFDA	download.open.fda.gov/drug/ndc/	137733 recs	Product NDC, labeler, route, dosage form, application number
FDA Drug Recalls	openFDA	download.open.fda.gov/drug/enforcement/	17899 recs	Recall classification, reason, initiation

¹ <https://www.vizient.com/newsroom/news-releases/new-vizient-survey-finds-drug-shortages-cost-hospitals-nearly-900m-annually-in-labor-expenses>

				date, recalling firm
Orange Book	FDA	fda.gov/media/76860/download	48663 products	Ingredient, dosage form/route, application type, approval date
Medicare Part B / D Spending	CMS	data.cms.gov/data-api/v1/dataset/	14536 rows	Spending, claims, beneficiaries, manufacturer count

Access: All sources are US federal public-domain data available as bulk downloads with no login or API key; the archived shortage files come from the Internet Archive. Total download is about 100 MB. The NDC product code would be the best join key: it appears on all shortage records and matches the NDC directory exactly for most of them.

Known constraints: The live FDA shortage file lists only active shortages; resolved ones are deleted after about six months, leaving just 7 in the current file. Shortage history has to come from archived copies instead. Recalls carry a structured drug code on only 17.9% of rows; matching on ingredient name raises that to 60%. openFDA limits anonymous use to 1,000 requests per day, so we use the bulk files rather than the API.

2. Feasibility, Bias, Ethics, and Limitations

- **Incomplete shortage history:** Because FDA deletes resolved shortages, the live file over-represents long-running ones and would give us only about 70 biased training examples. **Mitigation:** we reconstruct history from 102 archived snapshots, yielding 10,967 shortage records and 100–230 new shortages per year versus the 3–17 visible live.
- **Fields that give away the answer:** Anything derived from how long a record has been posted predicts the label almost perfectly while being useless in practice. **Mitigation:** we will fix a banned-feature list before modeling and report precision-recall against the base rate.
- **Time-dependent structure:** Shortages cluster in time and their causes shift: FDA attributes most 2013–17 shortages to manufacturing quality but most 2023 shortages to demand. **Mitigation:** train on earlier years and test on later ones rather than splitting randomly, and report results by period instead of pooling them.